

Lemma: $b \in \mathbb{R}$ 且 $b \geq 2$, 则对 $\forall k \in \mathbb{Z}_{\geq 0}$, 都有 $b^k \geq k+1 > k$

Proof: 对于 $k=0$, $b^k = b^0 = 1 \geq 0+1 > 0 \quad \therefore b^k \geq k+1 > k$

对于 $k=1$, $b^k = b^1 = b \geq 2 \geq 1+1 > 1 \quad \therefore b^k \geq k+1 > k$

对于 $k=2$, $\because b \geq 2 \quad \therefore b^2 \geq 2^2 = 4$

$\therefore b^k = b^2 \geq 4 > 2+1 > 2 \quad \therefore b^k \geq k+1 > k$

假设 $k=\lambda$ ($\lambda \in \mathbb{Z}_{\geq 2}$) 时, 结论 ~~成立~~ 成立.

则当 $k=\lambda+1$ 时, $b^k = b^{\lambda+1} = b^\lambda \cdot b$.

$\therefore b^\lambda \geq \lambda+1 > \lambda$, $b \geq 2 > 0$.

$\therefore b^\lambda \cdot b \geq (\lambda+1)b \quad \therefore b^k = b^\lambda \cdot b \geq (\lambda+1)b$

$\because b \geq 2, \lambda+1 > 0 \quad \therefore (\lambda+1)b \geq 2(\lambda+1) ~~\geq \lambda+1~~$

$\therefore (\lambda+1)b \geq 2\lambda+2 \geq \lambda+2 > \lambda+1$

$\therefore b^k \geq (\lambda+1)b \geq \lambda+2 > \lambda+1$

$\therefore b^k \geq k+1 > k \quad \therefore$ 当 $k=\lambda+1$ 时结论成立.

$\therefore$ 对 $\forall k \in \mathbb{Z}_{\geq 0}$, 都有 $b^k \geq k+1 > k$. $\square$

Lemma: 设 $(N, 0, s)$ 满足 Peano 公理, $x, y \in N$. 若 $x \leq s(y)$ 且 $x \neq s(y)$, 则 $x \leq y$.

Proof: $\because y \in N$, $s: N \rightarrow N$ 是一个映射 $\therefore s(y) \in N$

$\because x \in N, s(y) \in N, x \leq s(y) \quad \therefore \exists d \in N, \text{ s.t. } s(y) = x+d$.

假设 $d=0$, 则有 $s(y) = x+d = x+0 = x \quad \therefore x = s(y)$ 矛盾. $\therefore d \neq 0$

$$\because d \in \mathbb{N} \text{ 且 } d \neq 0 \quad \therefore \exists \alpha \in \mathbb{N} \text{ s.t. } d \neq \alpha \text{ 且 } d = s(\alpha)$$

$$\therefore s(y) = x + d = x + s(\alpha) = s(x + \alpha) \quad \therefore \text{映射 } s: \mathbb{N} \rightarrow \mathbb{N} \text{ 是单射}$$

$$\therefore y = x + \alpha, \alpha \in \mathbb{N} \quad \therefore x \leq y \quad \square$$

定理(最大自然数原理) 设 $(\mathbb{N}, 0, s)$ 满足 Peano 公理, A 是 $\mathbb{N}$ 的一个任意的非空且有上界的子集, 则 A 中一定存在最大的元素.

Proof: 已经证明了 $(\mathbb{N}, \leq)$ 是全序集. $\therefore A$ 是 $\mathbb{N}$ 的非空子集.

$\therefore (A, \leq)$ 也是全序集.

$$\text{令 } A^b = \{x \in \mathbb{N} \mid \exists y \in A, \text{ s.t. } x \leq y\} \quad \therefore A^b \subseteq \mathbb{N}$$

$$\because A \neq \emptyset \quad \therefore \text{从 } A \text{ 中任取一个元素, 记为 } \lambda \quad \therefore \lambda \in A \subseteq \mathbb{N} \quad \therefore \lambda \in \mathbb{N}$$

$$\because 0 \in \mathbb{N} \text{ 且 } \lambda \in \mathbb{N} \text{ 且 } \lambda = \lambda + 0 = 0 + \lambda \quad \therefore 0 \leq \lambda$$

$$\because 0 \in \mathbb{N}, \lambda \in A, 0 \leq \lambda \quad \therefore 0 \in A^b \quad \therefore A^b \text{ 是 } \mathbb{N} \text{ 的包含 } 0 \text{ 的子集.}$$

$$\because A \text{ 有上界} \quad \therefore \exists M \in \mathbb{N}, \text{ s.t. 对 } \forall y \in A, \text{ 都有 } y \leq M$$

$$\text{假设 } s(M) \in A^b \quad \therefore M \in \mathbb{N}, s: \mathbb{N} \rightarrow \mathbb{N} \text{ 是一个映射} \quad \therefore s(M) \in \mathbb{N}$$

$$\text{假设 } s(M) \in A^b, \text{ 则 } \exists \beta \in A, \text{ s.t. } s(M) \leq \beta \quad \therefore \beta \in A \quad \therefore \beta \leq M$$

$$\because M \in \mathbb{N} \quad \therefore M < s(M) \quad \therefore s(M) \leq \beta \leq M < s(M) \text{ 矛盾. } \therefore s(M) \notin A^b$$

$$\because s(M) \in \mathbb{N} \text{ 且 } s(M) \notin A^b \text{ 且 } A^b \subseteq \mathbb{N} \quad \therefore A^b \subsetneq \mathbb{N}$$

$$\therefore \exists \zeta \in \mathbb{N}, \text{ s.t. } \zeta \in A^b \text{ 但是 } s(\zeta) \notin A^b \quad \therefore \zeta \in \mathbb{N} \quad \therefore s(\zeta) \in \mathbb{N}$$

$$\text{对 } \forall y \in A, \quad \therefore s(\zeta) \notin A^b \quad \therefore s(\zeta) \not\leq y$$

$$\because s(\zeta) \in \mathbb{N}, y \in A \subseteq \mathbb{N}, s(\zeta) \not\leq y, (\mathbb{N}, \leq) \text{ 是全序集} \quad \therefore y < s(\zeta)$$

$\therefore y \leq \zeta \quad \therefore \zeta$ 是 A 的一个上界.

$\therefore \zeta \in A^b \quad \therefore \exists d \in A, \text{ s.t. } \zeta \leq d \quad \therefore d \in A, \zeta$ 是 A 的一个上界

$\therefore d \leq \zeta \quad \therefore \zeta \leq d \leq \zeta \quad \therefore \zeta = d \in A \quad \therefore \zeta \in A$

$\therefore \zeta$ 是 A 的最大值. ζ 是 A 中最大的元素. $\square$

定义: p 是素数, $a \in \mathbb{Z}_{\geq 0}$, $n \in \mathbb{Z} \setminus \{0\}$. 若 $p^a | n$ 且 $p^{a+1} \nmid n$, 则记 $p^a \parallel n$.

Lemma (整数环 $\mathbb{Z}$ 上的 p -进赋值) p 是一个任意的素数, m 是一个任意的非零整数, 则有: 存在唯一的 $v_p(m) \in \mathbb{Z}_{\geq 0}$, 使得

$$p^{v_p(m)} \parallel m$$

Proof: 定义集合 $S = \{k \in \mathbb{Z}_{\geq 0} \mid p^k | m\}$ $\therefore S \subseteq \mathbb{Z}_{\geq 0}$

$\because 0 \in \mathbb{Z}_{\geq 0}$ 且 $p^0 | m$ $\therefore 0 \in S$ $\therefore S \subseteq \mathbb{Z}_{\geq 0}$ 且 $S \neq \emptyset$

$\because m \in \mathbb{Z} \setminus \{0\}$ $\therefore |m| \in \mathbb{Z}_{\geq 1}$

对 $\forall k \in S$, 有: $k \in \mathbb{Z}_{\geq 0}$ 且 $p^k | m$ $\therefore \exists q \in \mathbb{Z}$, s.t. $m = p^k \cdot q$

假设 $q = 0$, 则 $m = p^k \cdot q = p^k \cdot 0 = 0$. 矛盾. $\therefore q \neq 0$ $\therefore q \in \mathbb{Z} \setminus \{0\}$.

$\therefore |q| \in \mathbb{Z}_{\geq 1}$ $\therefore |m| = |p^k \cdot q| = \cancel{p^k} \cdot \cancel{q} = |p^k| \cdot |q| = p^k \cdot |q|$

$\because |q| \geq 1$, $p^k \geq 1 > 0$ $\therefore p^k \cdot |q| \geq p^k$ $\therefore |m| \geq p^k$

$\because p$ 是素数 $\therefore p \in \mathbb{Z}_{\geq 2}$ $\therefore k \in \mathbb{Z}_{\geq 0}$ $\therefore p^k \geq k+1 > k$

$\therefore k < p^k \leq |m|$ $\therefore k < |m|$

$\therefore |m| \in \mathbb{Z}_{\geq 1}$ 且 $|m|$ 是集合 S 的一个上界

$\therefore S$ 中存在最大的数. 设 $\gamma = \max S$ $\therefore \gamma+1 \notin S$ $\therefore \gamma \in S$

$$\therefore \gamma \in \mathbb{Z}_{\geq 0} \text{ 且 } p^\gamma \mid m \text{ 且 } \gamma+1 \in \mathbb{Z}_{\geq 1} \text{ 且 } p^{\gamma+1} \nmid m$$

$$\therefore \gamma \in \mathbb{Z}_{\geq 0} \text{ 且 } p^\gamma \mid m \text{ 且 } p^{\gamma+1} \nmid m \quad \therefore p^\gamma \parallel m \quad \text{存在性得证.}$$

$$\text{假设存在 } \delta, \varepsilon \in \mathbb{Z}_{\geq 0}, \text{ 使得 } p^\delta \parallel m \text{ 且 } p^\varepsilon \parallel m$$

$$\therefore p^\delta \parallel m \quad \therefore p^\delta \mid m \text{ 且 } p^{\delta+1} \nmid m$$

$$\therefore p^\varepsilon \parallel m \quad \therefore p^\varepsilon \mid m \text{ 且 } p^{\varepsilon+1} \nmid m.$$

$$\text{如果 } \delta < \varepsilon, \text{ 则 } \delta+1 \leq \varepsilon \quad \therefore p^{\delta+1} \mid p^\varepsilon$$

$$\therefore p^{\delta+1} \mid p^\varepsilon, p^\varepsilon \mid m \quad \therefore p^{\delta+1} \mid m. \text{ 矛盾.}$$

$$\text{如果 } \delta > \varepsilon, \text{ 则 } \delta \geq \varepsilon+1 \quad \therefore p^{\varepsilon+1} \mid p^\delta$$

$$\therefore p^{\varepsilon+1} \mid p^\delta, p^\delta \mid m \quad \therefore p^{\varepsilon+1} \mid m. \text{ 矛盾} \quad \therefore \delta = \varepsilon. \text{ 唯一性得证.}$$

$$\therefore \text{存在唯一的非负整数 } v_p(m), \text{ 使得 } p^{v_p(m)} \parallel m.$$

$$\therefore \text{存在唯一的 } v_p(m) \in \mathbb{Z}_{\geq 0}, \text{ s.t. } p^{v_p(m)} \parallel m$$

$$\therefore v_p \text{ 可以看成 } \mathbb{Z} \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0} \text{ 的映射.} \quad \square$$

Lemma (p -进赋值值的对数性质) p 是素数, $x, y \in \mathbb{Z} \setminus \{0\}$,

$$\text{则有: } v_p(xy) = v_p(x) + v_p(y)$$

$$\text{Proof: } \therefore p \text{ 是素数, } x \in \mathbb{Z} \setminus \{0\} \quad \therefore \text{存在唯一的 } v_p(x) \in \mathbb{Z}_{\geq 0}, \text{ 使得}$$

$$p^{v_p(x)} \parallel x. \text{ 记 } a = v_p(x) \in \mathbb{Z}_{\geq 0} \quad \therefore a \in \mathbb{Z}_{\geq 0} \text{ 且 } p^a \parallel x.$$

$\because p$ 是素数, $y \in \mathbb{Z} \setminus \{0\}$ $\therefore$ 存在唯一的 $v_p(y) \in \mathbb{Z}_{\geq 0}$, 使得
 $p^{v_p(y)} \parallel y$. 记 $b = v_p(y) \in \mathbb{Z}_{\geq 0}$ $\therefore b \in \mathbb{Z}_{\geq 0}$ 且 $p^b \parallel y$

$\because x, y \in \mathbb{Z} \setminus \{0\}$ $\therefore xy \in \mathbb{Z} \setminus \{0\}$

$\because p$ 是素数, $xy \in \mathbb{Z} \setminus \{0\}$ $\therefore$ 存在唯一的 $v_p(xy) \in \mathbb{Z}_{\geq 0}$, s.t.
 $p^{v_p(xy)} \parallel xy$. 记 $c = v_p(xy) \in \mathbb{Z}_{\geq 0}$ $\therefore c \in \mathbb{Z}_{\geq 0}$ 且 $p^c \parallel xy$

$\because p^a \parallel x$ $\therefore p^a \mid x$ 且 $p^{a+1} \nmid x$

$\because p^a \mid x$ $\therefore \exists q_1 \in \mathbb{Z}$, s.t. $x = p^a \cdot q_1$.

假设 $p \mid q_1$, 则 $\exists k_1 \in \mathbb{Z}$, s.t. $q_1 = p k_1$

$\therefore \cancel{x = p^a \cdot q_1} x = p^a \cdot q_1 = p^a \cdot p k_1 = p^{a+1} \cdot k_1$ $\therefore p^{a+1} \mid x$ 矛盾.

$\therefore p \nmid q_1$

$\because p^b \parallel y$ $\therefore p^b \mid y$ 且 $p^{b+1} \nmid y$

$\because p^b \mid y$ $\therefore \exists q_2 \in \mathbb{Z}$, s.t. $y = p^b \cdot q_2$

假设 $p \mid q_2$, 则 $\exists k_2 \in \mathbb{Z}$, s.t. $q_2 = p k_2$

$\therefore y = p^b q_2 = p^b \cdot p \cdot k_2 = p^{b+1} \cdot k_2$ $\therefore p^{b+1} \mid y$ 矛盾.

$\therefore p \nmid q_2$

$\therefore x = p^a \cdot q_1, q_1 \in \mathbb{Z}, y = p^b \cdot q_2, q_2 \in \mathbb{Z}$

$$\therefore xy = (p^a \cdot q_1)(p^b \cdot q_2) = p^a \cdot p^b \cdot q_1 \cdot q_2 = p^{a+b} \cdot (q_1 q_2)$$

$$\therefore \frac{xy}{p^{a+b}} = q_1 q_2 \in \mathbb{Z} \quad \therefore p^{a+b} \mid xy$$

假设 $p^{a+b+1} \mid xy$, 则 $\exists k_3 \in \mathbb{Z}$, s.t. $xy = p^{a+b+1} \cdot k_3$

$$\therefore \cancel{p^{a+b}} p^{a+b} \cdot (q_1 q_2) = xy = p^{a+b+1} \cdot k_3 = p^{a+b} \cdot p k_3 \quad \therefore p^{a+b} \geq p^0 = 1 > 0$$

$$\therefore q_1 q_2 = p k_3 \quad \therefore p \mid q_1 q_2$$

$\therefore p$ 是素数, $q_1 \in \mathbb{Z}$, $q_2 \in \mathbb{Z}$, $p \mid q_1 q_2 \quad \therefore p \mid q_1$ 或 $p \mid q_2$

$\therefore p \nmid q_1$ 且 $p \nmid q_2 \quad \therefore$ 矛盾. $\therefore p^{a+b+1} \nmid xy$.

$\therefore p$ 是素数, $a+b \in \mathbb{Z}_{\geq 0}$, $xy \in \mathbb{Z} \setminus \{0\}$, $p^{a+b} \mid xy$ 且 $p^{a+b+1} \nmid xy$

$$\therefore p^{a+b} \parallel xy$$

$\therefore a+b \in \mathbb{Z}_{\geq 0}$ 且 $p^{a+b} \parallel xy$, 存在唯一的 $v_p(xy) \in \mathbb{Z}_{\geq 0}$ s.t.

$$p^{v_p(xy)} \parallel xy \quad \therefore v_p(xy) = a+b = v_p(x) + v_p(y) \quad \square$$

~~例 (计算具体的 p 进赋值)~~

Lemma: p 是素数, 对 $\forall m \in \mathbb{Z} \setminus \{0\}$, 有: $v_p(-m) = v_p(m)$

Proof: $\equiv p$ 是素数, 对 $\forall m \in \mathbb{Z} \setminus \{0\}$, 有:

$\therefore p$ 是素数, $m \in \mathbb{Z} \setminus \{0\} \quad \therefore$ 存在唯一的 $v_p(m) \in \mathbb{Z}_{\geq 0}$, 使得 $p^{v_p(m)} \parallel m$

记 $a = v_p(m) \in \mathbb{Z}_{\geq 0}$, 则有 $a \in \mathbb{Z}_{\geq 0}$ 且 $p^a \parallel m$.

$\because m \in \mathbb{Z} \setminus \{0\} \quad \therefore -m \in \mathbb{Z} \setminus \{0\} \quad \because p \text{ 是素数}, -m \in \mathbb{Z} \setminus \{0\}$

$\therefore$ 存在唯一的 $v_p(-m) \in \mathbb{Z}_{\geq 0}$, s.t. $p^{v_p(-m)} \parallel -m$

$\because p^a \parallel m \quad \therefore p^a \mid m$ 且 $p^{a+1} \nmid m$

$\because p^a \mid m \quad \therefore \exists q \in \mathbb{Z}$, s.t. $m = p^a \cdot q$

$\therefore -m = -p^a \cdot q = p^a \cdot (-q)$ 且 $-q \in \mathbb{Z} \quad \therefore p^a \mid -m$.

假设 $p^{a+1} \mid -m$. 则 $\exists k \in \mathbb{Z}$, s.t. $-m = p^{a+1} \cdot k$

$\therefore m = -p^{a+1} \cdot k = p^{a+1} \cdot (-k)$, 其中 $-k \in \mathbb{Z} \quad \therefore p^{a+1} \mid m$ 矛盾.

$\therefore p^{a+1} \nmid -m$

$\because p \text{ 是素数}, a \in \mathbb{Z}_{\geq 0}, -m \in \mathbb{Z} \setminus \{0\}, p^a \mid -m$ 且 $p^{a+1} \nmid -m$

$\therefore p^a \parallel -m. \quad \therefore$ 存在唯一的 $v_p(-m) \in \mathbb{Z}_{\geq 0}$, s.t. $p^{v_p(-m)} \parallel -m$

$\therefore v_p(-m) = a = v_p(m) \quad \square$

例 (具体计算 $v_p(n)$)

1. 对任意素数 p , 都有 $v_p(1) = 0$.

$\because p \text{ 是素数}, 1 \in \mathbb{Z} \setminus \{0\}. \quad \therefore$ 存在唯一的 $v_p(1) \in \mathbb{Z}_{\geq 0}$, 使得 $p^{v_p(1)} \parallel 1$

$\because p \text{ 是素数}, 0 \in \mathbb{Z}_{\geq 0}, 1 \in \mathbb{Z} \setminus \{0\}, p^0 \mid 1$ 且 $p^1 \nmid 1 \quad \therefore p^0 \parallel 1$

$\therefore v_p(1) = 0$

$$2. \quad V_2(2) = 1, \quad \text{当素数 } p \neq 2 \text{ 时}, \quad V_p(2) = 0$$

$$\because 2^1 \mid 2 \text{ 且 } 2^2 \nmid 2 \quad \therefore 2^1 \parallel 2 \quad \therefore V_2(2) = 1$$

$$\text{当素数 } p \neq 2 \text{ 时}, \quad \because p^0 \mid 2 \text{ 且 } p^1 \nmid 2 \quad \therefore p^0 \parallel 2 \quad \therefore V_p(2) = 0$$

Lemma: p 是素数, $n \in \mathbb{Z} \setminus \{0\}$. 则有: $p \nmid n \Leftrightarrow V_p(n) = 0$

$$\text{proof: } (\Rightarrow): \because p \nmid n \quad \therefore p^0 \mid n \text{ 且 } p^1 \nmid n \quad \therefore p^0 \parallel n \quad \therefore V_p(n) = 0$$

$$(\Leftarrow): \because p^{V_p(n)} \parallel n \quad \therefore p^0 \parallel n \quad \therefore p^0 \mid n \text{ 且 } p^1 \nmid n \quad \therefore p \nmid n$$

Lemma: p 是素数, $n \in \mathbb{Z}_{\geq 1}$. 定义集合

$$X := \{ (m, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1} \mid j \leq V_p(m) \}$$

$$Y := \{ (j, q) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1} \mid q \leq \left\lfloor \frac{n}{p^j} \right\rfloor \}$$

则有: ① 集合 X 和 Y 良定义

$$\text{② 当 } n < p \text{ 时}, \quad X = \emptyset \text{ 且 } Y = \emptyset$$

$$\text{③ 当 } n \geq p \text{ 时}, \quad X \neq \emptyset \text{ 且 } Y \neq \emptyset$$

$$\text{④ } |X| = |Y|$$

proof: ①: 对 $\forall (m, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1}$, 有: $m \in \{1, 2, \dots, n\}$ 且 $j \in \mathbb{Z}_{\geq 1}$

$$\because m \in \mathbb{Z} \setminus \{0\}. \quad \therefore \text{存在唯一的 } V_p(m) \in \mathbb{Z}_{\geq 0}, \text{ s.t. } p^{V_p(m)} \parallel m.$$

$$\because j \in \mathbb{Z}_{\geq 1} \text{ 且 } V_p(m) \in \mathbb{Z}_{\geq 0} \quad \therefore j \leq V_p(m) \text{ 必然能返回一个确定的真或假.}$$

∴ 集合 X 良定义.

对 $\forall (j, q) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1}$, 有: $j \in \mathbb{Z}_{\geq 1}$ 且 $q \in \mathbb{Z}_{\geq 1}$ ∴ $p^j \geq p' = p \geq 2$

$$\therefore p^j \in \mathbb{Z}_{\geq 2} \quad \therefore \frac{n}{p^j} \in \mathbb{R}_{>0} \quad \therefore 0 \leq \frac{n}{p^j} \quad \therefore 0 \leq \left\lfloor \frac{n}{p^j} \right\rfloor$$

$$\therefore \left\lfloor \frac{n}{p^j} \right\rfloor \in \mathbb{Z}_{\geq 0} \quad \therefore q \in \mathbb{Z}_{\geq 1} \quad \therefore q \leq \left\lfloor \frac{n}{p^j} \right\rfloor \text{ 必然能返回一个确定的真或假.}$$

∴ 集合 Y 良定义. ① 得证.

当 $n < p$ 时,

假设 $X \neq \emptyset$. 则 $\exists (m, j) \in X$ ∴ $m \in \{1, 2, \dots, n\}$ 且 $j \in \mathbb{Z}_{\geq 1}$ 且 $j \leq v_p(m)$

$$\therefore n < p \quad \therefore 1 \leq m \leq n < p \quad \therefore p \nmid m \quad \therefore v_p(m) = 0$$

$$\therefore j \geq 1 \text{ 且 } j \leq 0 \text{ 矛盾} \quad \therefore X = \emptyset$$

假设 $Y \neq \emptyset$. 则 $\exists (j, q) \in Y$ ∴ $j \in \mathbb{Z}_{\geq 1}$ 且 $q \in \mathbb{Z}_{\geq 1}$ 且 $q \leq \left\lfloor \frac{n}{p^j} \right\rfloor$

$$\therefore j \geq 1 \text{ 且 } p \in \mathbb{Z}_{\geq 2} \quad \therefore p^j \geq p' = p \geq 2 \quad \therefore 0 < n < p \leq p^j$$

$$\therefore 0 < \frac{n}{p^j} < \frac{p}{p^j} \leq 1 \quad \therefore 0 < \frac{n}{p^j} < 1 \quad \therefore \left\lfloor \frac{n}{p^j} \right\rfloor = 0$$

$$\therefore q \geq 1 \text{ 且 } q \leq 0 \text{ 矛盾.} \quad \therefore Y = \emptyset \quad \text{② 得证.}$$

当 $n \geq p$ 时,

$$\therefore p \in \mathbb{Z}_{\geq 2} \text{ 且 } n \in \mathbb{Z}_{\geq 1} \text{ 且 } n \geq p \quad \therefore p \in \{1, 2, \dots, n\}, 1 \in \mathbb{Z}_{\geq 1}$$

$$\therefore p' \mid p \text{ 且 } p^2 \nmid p \quad \therefore p' \parallel p \quad \therefore v_p(p) = 1 \quad \therefore 1 \leq v_p(p)$$

$$\therefore (p, 1) \in X \quad \therefore X \neq \emptyset.$$

$$\because n \geq p \geq 2 > 1 > 0 \quad \therefore \frac{n}{p} \geq 1 \quad \therefore 1 \leq \frac{n}{p} \quad \therefore 1 \leq \left\lfloor \frac{n}{p} \right\rfloor$$

$$\therefore (1, 1) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1} \text{ 且 } 1 \leq \left\lfloor \frac{n}{p^1} \right\rfloor \quad \therefore (1, 1) \in Y \quad \therefore Y \neq \emptyset.$$

③得证.

$$\text{定义映射 } \Phi: X \longrightarrow Y \quad (\text{以下认为 } n \geq p)$$

$$(m, j) \longmapsto (j, \frac{m}{p^j})$$

$$\text{对 } \forall (m, j) \in X, \text{ 有: } m \in \{1, 2, \dots, n\} \text{ 且 } j \in \mathbb{Z}_{\geq 1} \text{ 且 } j \leq v_p(m)$$

$$\because p \text{ 是素数, } m \in \mathbb{Z} \setminus \{0\} \quad \therefore \text{存在唯一的 } v_p(m) \in \mathbb{Z}_{\geq 0} \text{ s.t. } p^{v_p(m)} \parallel m$$

$$\because p^{v_p(m)} \parallel m \quad \therefore p^{v_p(m)} \mid m \text{ 且 } p^{v_p(m)+1} \nmid m \quad \because j \in \mathbb{Z}_{\geq 1} \text{ 且 } j \leq v_p(m)$$

$$\therefore p^j \mid p^{v_p(m)} \quad \therefore p^j \mid m \quad \therefore \frac{m}{p^j} \in \mathbb{Z}_{\geq 1} \quad \therefore (j, \frac{m}{p^j}) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1}$$

$$\because m \leq n, \quad \cancel{p^j \in \mathbb{Z}} \quad p^j \geq p^1 = p \geq 2 > 1 > 0 \quad \therefore \frac{m}{p^j} \leq \frac{n}{p^j}$$

$$\therefore \frac{m}{p^j} \leq \frac{n}{p^j} \text{ 且 } \frac{m}{p^j} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{m}{p^j} \leq \left\lfloor \frac{n}{p^j} \right\rfloor$$

$$\therefore \Phi(m, j) = (j, \frac{m}{p^j}) \in Y \quad \therefore \Phi(X) \subseteq Y.$$

$$\text{对 } \forall (m_1, j_1), (m_2, j_2) \in X.$$

$$\text{若 } (m_1, j_1) = (m_2, j_2), \text{ 则 } m_1 = m_2 \text{ 且 } j_1 = j_2$$

$$\therefore m_1 = m_2 \in \{1, 2, \dots, n\} \text{ 且 } j_1 = j_2 \in \mathbb{Z}_{\geq 1} \quad \therefore p^{j_1} = p^{j_2}$$

$$\therefore \Phi(m_1, j_1) = (j_1, \frac{m_1}{p^{j_1}}) = (j_2, \frac{m_2}{p^{j_2}}) = \Phi(m_2, j_2) \quad \therefore \Phi \text{ 是一个映射.}$$

$$\text{若 } \Phi(m_1, j_1) = \Phi(m_2, j_2), \text{ 则 } (j_1, \frac{m_1}{p^{j_1}}) = (j_2, \frac{m_2}{p^{j_2}})$$

$$\therefore j_1 = j_2 \text{ 且 } \frac{m_1}{p^{j_1}} = \frac{m_2}{p^{j_2}} \quad \therefore j_1 = j_2 \quad \therefore p^{j_1} = p^{j_2}$$

$$\therefore j_1 = j_2 \in \mathbb{Z}_{\geq 1} \quad \therefore p^{j_1} = p^{j_2} \in \mathbb{Z}_{\geq 1} \quad \therefore m_1 = m_2$$

$$\therefore (m_1, j_1) = (m_2, j_2) \quad \therefore \Phi \text{ 是一个单射.}$$

$$\text{对 } \forall (j, q) \in Y, \text{ 有: } j \in \mathbb{Z}_{\geq 1} \text{ 且 } q \in \mathbb{Z}_{\geq 1} \text{ 且 } q \leq \lfloor \frac{n}{p^j} \rfloor$$

$$\text{考虑 } (qp^j, j) \quad \therefore j \in \mathbb{Z}_{\geq 1} \quad \therefore j \in \mathbb{Z}_{\geq 1}$$

$$\therefore p \in \mathbb{Z}_{\geq 2} \text{ 且 } j \in \mathbb{Z}_{\geq 1} \quad \therefore p^j \geq p^1 = p \geq 2 > 1 > 0 \quad \therefore p^j \in \mathbb{Z}_{\geq 2}$$

$$\therefore q \in \mathbb{Z}_{\geq 1} \quad \therefore qp^j \in \mathbb{Z}_{\geq 1}$$

$$\therefore q \in \mathbb{Z}_{\geq 1} \text{ 且 } q \leq \lfloor \frac{n}{p^j} \rfloor \leq \frac{n}{p^j} \text{ 且 } p^j \in \mathbb{Z}_{\geq 1} \quad \therefore qp^j \leq n$$

$$\therefore qp^j \in \{1, 2, \dots, n\} \quad \therefore (qp^j, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1}$$

$$\therefore p \text{ 是素数, } qp^j \in \mathbb{Z}_{\geq 1} \quad \therefore \text{存在唯一的 } v_p(qp^j) \in \mathbb{Z}_{\geq 0}, \text{ s.t.}$$

$$p^{v_p(qp^j)} \parallel qp^j \quad \therefore p^{v_p(qp^j)} \mid qp^j \text{ 且 } p^{v_p(qp^j)+1} \nmid qp^j$$

$$\therefore p^j \in \mathbb{Z}_{\geq 1} \text{ 且 } q \in \mathbb{Z}_{\geq 1} \quad \therefore p^j \mid qp^j \quad \text{假设 } j > v_p(qp^j)$$

$$\text{则有 } \therefore j \in \mathbb{Z}_{\geq 1} \text{ 且 } v_p(qp^j) \in \mathbb{Z}_{\geq 0} \quad \therefore j \geq v_p(qp^j) + 1$$

$$\therefore j \geq v_p(qp^j) + 1 \text{ 且 } v_p(qp^j) + 1 \in \mathbb{Z}_{\geq 1} \quad \therefore p^{v_p(qp^j)+1} \mid p^j$$

$$\therefore p^{v_p(qp^j)+1} \mid qp^j \text{ 矛盾.} \quad \therefore j \leq v_p(qp^j)$$

$$\therefore (q p^j, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1} \text{ 且 } j \leq v_p(q p^j)$$

$$\therefore (q p^j, j) \in X \quad \text{至} (q p^j, j) = (j, \frac{q p^j}{p^j}) = (j, q)$$

$$\therefore \text{至是一个满射} \quad \therefore \text{至是一个双射} \quad \therefore |X| = |Y| \quad \textcircled{4} \text{得证. } \square$$

Lemma: p 是素数, $n \in \mathbb{Z}_{\geq 1}$, 定义集合

$$X = \{(m, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1} \mid j \leq v_p(m)\}$$

$$Y = \{(j, q) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1} \mid q \leq \lfloor \frac{n}{p^j} \rfloor\}$$

对 $\forall m \in \{1, 2, \dots, n\}$, 定义集合 $X_m = \{(m, j) \mid j \in \mathbb{Z}_{\geq 1} \text{ 且 } j \leq v_p(m)\}$

对 $\forall j \in \mathbb{Z}_{\geq 1}$, 定义集合 $Y_j = \{(j, q) \mid q \in \mathbb{Z}_{\geq 1} \text{ 且 } q \leq \lfloor \frac{n}{p^j} \rfloor\}$

$$\text{则有: } X = \bigsqcup_{m=1}^n X_m \text{ 且 } Y = \bigsqcup_{j=1}^{+\infty} Y_j$$

Proof: 对 $\forall (m, j) \in X$, 有: $m \in \{1, 2, \dots, n\}$ 且 $j \in \mathbb{Z}_{\geq 1}$ 且 $j \leq v_p(m)$

$$\therefore m \in \{1, 2, \dots, n\} \text{ 且 } (m, j) \in X_m \quad \therefore (m, j) \in \bigcup_{t=1}^n X_t = \text{右边}$$

$$\therefore \text{左边} \subseteq \text{右边}.$$

对 $\forall m \in \{1, 2, \dots, n\}$, 任取 X_m 中的一元 (m, j) (其中 $j \in \mathbb{Z}_{\geq 1}$ 且 $j \leq v_p(m)$)

$$\therefore (m, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1} \text{ 且 } j \leq v_p(m)$$

$$\therefore (m, j) \in X \quad \therefore X_m \subseteq X \quad \therefore \bigcup_{m=1}^n X_m \subseteq X \quad \therefore \text{右边} \subseteq \text{左边}$$

$$\therefore \text{左边} = \text{右边}.$$

对 $\forall m_1, m_2 \in \{1, 2, \dots, n\}$ 且 $m_1 \neq m_2$. 假设 $X_{m_1} \cap X_{m_2} \neq \emptyset$.

则 $\exists (\alpha, \beta) \in X_{m_1} \cap X_{m_2}$. $\because (\alpha, \beta) \in X_{m_1} \quad \therefore \alpha = m_1$

$\because (\alpha, \beta) \in X_{m_2} \quad \therefore \alpha = m_2 \quad \therefore m_1 = \alpha = m_2$. 矛盾. $\therefore X_{m_1} \cap X_{m_2} = \emptyset$

$$\therefore X = \bigsqcup_{m=1}^n X_m.$$

对 $\forall (j, q) \in Y$, 有: $j \in \mathbb{Z}_{\geq 1}$ 且 $q \in \mathbb{Z}_{\geq 1}$ 且 $q \leq \lfloor \frac{n}{p^j} \rfloor$

$\because j \in \mathbb{Z}_{\geq 1}$ 且 $(j, q) \in Y_j \quad \therefore (j, q) \in \text{右边} \quad \therefore \text{左边} \subseteq \text{右边}$.

对 $\forall j \in \mathbb{Z}_{\geq 1}$, 任取 Y_j 中的一元 (j, q) (其中 $q \in \mathbb{Z}_{\geq 1}$ 且 $q \leq \lfloor \frac{n}{p^j} \rfloor$)

$\therefore (j, q) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1}$ 且 $q \leq \lfloor \frac{n}{p^j} \rfloor \quad \therefore (j, q) \in Y \quad \therefore Y_j \subseteq Y$

$\therefore \text{右边} \subseteq \text{左边} \quad \therefore \text{左边} = \text{右边}$.

对 $\forall j_1, j_2 \in \mathbb{Z}_{\geq 1}$ 且 $j_1 \neq j_2$. 假设 $Y_{j_1} \cap Y_{j_2} \neq \emptyset$, 则

$\exists (\alpha, \beta) \in Y_{j_1} \cap Y_{j_2} \quad \because (\alpha, \beta) \in Y_{j_1} \quad \therefore \alpha = j_1$

$\because (\alpha, \beta) \in Y_{j_2} \quad \therefore \alpha = j_2 \quad \therefore j_1 = \alpha = j_2$ 矛盾 $\therefore Y_{j_1} \cap Y_{j_2} = \emptyset$

$$\therefore Y = \bigsqcup_{j=1}^{+\infty} Y_j.$$

Remark: $\exists N \in \mathbb{Z}_{\geq 1}$, s.t. 对 $\forall j > N$, 都有 $p^j > n \geq 1$

$\therefore$ 对 $\forall j > N$, 都有 $0 < \frac{n}{p^j} < 1 \quad \therefore$ 对 $\forall j > N$, 都有 $\lfloor \frac{n}{p^j} \rfloor = 0$

$\therefore$ 对 $\forall j > N$, 都有 $Y_j = \emptyset$

$\therefore \exists N \in \mathbb{Z}_{\geq 1}$, s.t. 对 $\forall j > N$, 都有 $Y_j = \emptyset$. $\square$

定理 (Legendre's formula) p 是素数, $n \in \mathbb{Z}_{\geq 0}$, 则有:

$$v_p(n!) = \sum_{k=1}^{+\infty} \left\lfloor \frac{n}{p^k} \right\rfloor$$

Proof: 当 $n=0$ 时, 左边 $= v_p(0!) = v_p(1) = 0$

$$\text{右边} = \sum_{k=1}^{+\infty} \left\lfloor \frac{0}{p^k} \right\rfloor = \sum_{k=1}^{+\infty} 0 = 0 \quad \therefore \text{左边} = \text{右边}.$$

当 $n \in \mathbb{Z}_{\geq 1}$ 时, $\because p$ 是素数, $n \in \mathbb{Z}_{\geq 1} \quad \therefore$ 定义集合

$$X = \{(m, j) \in \{1, 2, \dots, n\} \times \mathbb{Z}_{\geq 1} \mid j \leq v_p(m)\}$$

$$Y = \{(j, q) \in \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1} \mid q \leq \left\lfloor \frac{n}{p^j} \right\rfloor\} \quad \text{则已经证明了 } |X| = |Y|.$$

对 $\forall m \in \{1, 2, \dots, n\}$, 定义集合 $X_m = \{(m, j) \mid j \in \mathbb{Z}_{\geq 1} \text{ 且 } j \leq v_p(m)\}$

$$\text{则已经证明了 } X = \bigsqcup_{m=1}^n X_m$$

对 $\forall j \in \mathbb{Z}_{\geq 1}$, 定义集合 $Y_j = \{(j, q) \mid q \in \mathbb{Z}_{\geq 1} \text{ 且 } q \leq \left\lfloor \frac{n}{p^j} \right\rfloor\}$

$$\text{则已经证明了 } Y = \bigsqcup_{j=1}^{+\infty} Y_j$$

$$\therefore |X| = \left| \bigsqcup_{m=1}^n X_m \right| = \sum_{m=1}^n |X_m| = \sum_{m=1}^n v_p(m) = v_p\left(\prod_{m=1}^n m\right) = v_p(n!)$$

$$|Y| = \left| \bigsqcup_{j=1}^{+\infty} Y_j \right| = \sum_{j=1}^{+\infty} |Y_j| = \sum_{j=1}^{+\infty} \left\lfloor \frac{n}{p^j} \right\rfloor = \sum_{k=1}^{+\infty} \left\lfloor \frac{n}{p^k} \right\rfloor$$

$$\therefore v_p(n!) = |X| = |Y| = \sum_{k=1}^{+\infty} \left\lfloor \frac{n}{p^k} \right\rfloor \quad \square$$

定理 (Legendre's formula, 另一种形式) p 是素数, $n \in \mathbb{Z}_{\geq 0}$.

$\because p$ 是素数 $\therefore p \in \mathbb{Z}_{\geq 2}$ $\because n \in \mathbb{Z}_{\geq 0}$ $\therefore n$ 有唯一的 p 进制展开

$$n = a_0 + a_1 p + a_2 p^2 + a_3 p^3 + \dots + a_r p^r \quad (\text{其中 } r \in \mathbb{Z}_{\geq 0}, a_i \in \{0, 1, \dots, p-1\})$$

$$\text{则有: } v_p(n!) = \frac{n - \sum_{i=0}^r a_i}{p-1}$$

Proof: 当 $n=0$ 时, 左边 $= v_p(0!) = v_p(1) = 0$

$$\because n=0 \therefore r=0 \text{ 且 } a_0=0 \therefore \text{右边} = \frac{n-a_0}{p-1} = \frac{0-0}{p-1} = 0$$

$\therefore$ 左边 = 右边.

当 $n \in \mathbb{Z}_{\geq 1}$ 时, 对 $\forall k \in \{1, 2, \dots, r\}$, 有:

$$\because n = \sum_{i=0}^r a_i p^i \therefore \frac{n}{p^k} = \frac{\sum_{i=0}^r a_i p^i}{p^k} = \sum_{i=0}^r a_i p^{i-k} = \sum_{i=0}^{k-1} a_i p^{i-k} + \sum_{i=k}^r a_i p^{i-k}$$

$$\therefore \frac{n}{p^k} = \sum_{i=k}^r a_i p^{i-k} + \sum_{i=0}^{k-1} a_i p^{i-k} = \sum_{i=k}^r a_i p^{i-k} + \frac{\sum_{i=0}^{k-1} a_i p^i}{p^k}$$

显然有 $\sum_{i=0}^{k-1} a_i p^i \geq 0$. 对 $\forall i=0, 1, \dots, k-1$, 都有 $0 \leq a_i \leq p-1$

$\therefore$ 对 $\forall i=0, 1, \dots, k-1$, 都有 $a_i p^i \leq (p-1)p^i$

$$\therefore \sum_{i=0}^{k-1} a_i p^i \leq \sum_{i=0}^{k-1} (p-1)p^i = (p-1) \sum_{i=0}^{k-1} p^i = (p-1) \cdot \frac{p^k - 1}{p-1} = p^k - 1 < p^k$$

$$\because p^k > 0 \therefore \frac{\sum_{i=0}^{k-1} a_i p^i}{p^k} < 1 \quad \because p^k > 0 \text{ 且 } 0 \leq \sum_{i=0}^{k-1} a_i p^i < p^k$$

$$\therefore 0 \leq \frac{\sum_{i=0}^{k-1} a_i p^i}{p^k} < 1 \quad \therefore \sum_{i=k}^r a_i p^{i-k} \in \mathbb{Z}_{\geq 0}$$

$$\therefore \left\lfloor \frac{n}{p^k} \right\rfloor = \left\lfloor \sum_{i=k}^r a_i p^{i-k} + \frac{\sum_{i=0}^{k-1} a_i p^i}{p^k} \right\rfloor = \sum_{i=k}^r a_i p^{i-k}$$

Remark: 当 $n \in \mathbb{Z}_{\geq 1}$ 且 $n < p$ 时, $n! = 1 \times 2 \times \dots \times n$, 显然有 $p \nmid n!$

$$\therefore n! \in \mathbb{Z}_{\geq 1} \text{ 且 } p \nmid n! \quad \therefore v_p(n!) = 0 \quad \therefore \text{左边} = 0$$

$$\therefore n \in \mathbb{Z}_{\geq 1} \text{ 且 } n < p \quad \therefore r = 0 \text{ 且 } n = a_0 \quad \therefore \text{右边} = \frac{n - \overset{a_0}{\cancel{a_0}}}{p-1} = 0 \quad \therefore \text{左边} = \text{右边}$$

当 $n \in \mathbb{Z}_{\geq 1}$ 且 $n \geq p$ 时, $r \in \mathbb{Z}_{\geq 1} \quad \therefore 1 \in \{1, 2, \dots, r\} \quad \therefore \{1, 2, \dots, r\} \neq \emptyset$

上述证明过程只需证明 $n \geq p$ 的情况即可.

Lemma: p 是素数, $x, y \in \mathbb{Z} \setminus \{0\}$, $y \mid x$, 则有:

$$v_p\left(\frac{x}{y}\right) = v_p(x) - v_p(y)$$

Proof: 设 $\lambda = \frac{x}{y}$. $\therefore x, y \in \mathbb{Z} \setminus \{0\}$ 且 $y \mid x \quad \therefore \lambda = \frac{x}{y} \in \mathbb{Z}$

$\therefore x \in \mathbb{Z} \setminus \{0\}$ 且 $y \in \mathbb{Z} \setminus \{0\} \quad \therefore \lambda = \frac{x}{y} \neq 0 \quad \therefore \lambda \in \mathbb{Z} \setminus \{0\}$

$\therefore p$ 是素数, $x \in \mathbb{Z} \setminus \{0\} \quad \therefore$ 存在唯一的 $v_p(x) \in \mathbb{Z}_{\geq 0}$ s.t. $p^{v_p(x)} \parallel x$

$\therefore p$ 是素数, $y \in \mathbb{Z} \setminus \{0\} \quad \therefore$ 存在唯一的 $v_p(y) \in \mathbb{Z}_{\geq 0}$ s.t. $p^{v_p(y)} \parallel y$

$\therefore p$ 是素数, $\lambda \in \mathbb{Z} \setminus \{0\} \quad \therefore$ 存在唯一的 $v_p(\lambda) \in \mathbb{Z}_{\geq 0}$ s.t. $p^{v_p(\lambda)} \parallel \lambda$

$\therefore \lambda = \frac{x}{y}, y \in \mathbb{Z} \setminus \{0\} \quad \therefore \lambda y = x, x = \lambda y$

$\therefore p$ 是素数, $\lambda \in \mathbb{Z} \setminus \{0\}, y \in \mathbb{Z} \setminus \{0\}$

$\therefore v_p(x) = v_p(\lambda y) = v_p(\lambda) + v_p(y) \quad \therefore v_p(x) - v_p(y) = v_p(\lambda)$

$\therefore v_p\left(\frac{x}{y}\right) = v_p(\lambda) = v_p(x) - v_p(y) \quad \square$

定义 (= 项式系数) $n \in \mathbb{Z}_{\geq 0}$, A 是一个任意的含有 n 个元素的有限集合. 对 $\forall k \in \mathbb{Z}$, 定义

$$\binom{n}{k} = \left| \{B \in P(A) \mid |B| = k\} \right| = \left| \{B \subseteq A \mid |B| = k\} \right|$$

~~Lemma: 对 $\forall k \in \mathbb{Z}$, 有: $\binom{0}{k} = 0$.~~

~~Proof: $\because A$ 是含有 0 个元素的集合 $\therefore A = \emptyset \therefore B \subseteq A \therefore B = \emptyset$.~~

~~$$\binom{0}{0} = 1$$~~

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 0}$, $\forall k \in \mathbb{Z}_{\leq -1}$ 或 $k \in \mathbb{Z}_{\geq n+1}$, 有:

$$\binom{n}{k} = 0.$$

Proof: A 是含有 n 个元素的有限集合. $B \subseteq A \therefore |B| \in \mathbb{Z}$ 且 $0 \leq |B| \leq n \therefore$ 当 $k \in \mathbb{Z}_{\leq -1}$ 或 $k \in \mathbb{Z}_{\geq n+1}$ 时, 有:

$$\{B \subseteq A \mid |B| = k\} = \emptyset \therefore \binom{n}{k} = |\emptyset| = 0 \quad \square$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 0}$, 有: $\binom{n}{0} = \binom{n}{n} = 1$.

$$\text{Proof: } \binom{n}{0} = \left| \{B \subseteq A \mid |B| = 0\} \right| = \left| \{\emptyset\} \right| = 1$$

$$\binom{n}{n} = \left| \{B \subseteq A \mid |B| = n\} \right| = \left| \{A\} \right| = 1 \quad \square$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 0}$, $\forall k \in \mathbb{Z}$, 有: $\binom{n}{k} \in \mathbb{Z}_{\geq 0}$.

Proof: 当 $k \in \mathbb{Z}_{\leq -1}$ 或 $k \in \mathbb{Z}_{\geq n+1}$ 时, 已证 $\binom{n}{k} = 0 \in \mathbb{Z}_{\geq 0}$.

已证 $\binom{n}{0} = \binom{n}{n} = 1 \in \mathbb{Z}_{\geq 0}$.

对 $\forall k \in \mathbb{Z}$ 且 $1 \leq k \leq n-1$, 有: $\because A$ 是含有 n 个元素的有限集合.

$\therefore P(A)$ 是含有 2^n 个元素的有限集合.

$\therefore \{B \subseteq A \mid |B|=k\} \subseteq P(A)$ 是有限集合.

$\therefore \binom{n}{k} = |\{B \subseteq A \mid |B|=k\}| \in \mathbb{Z}_{\geq 0}$. $\square$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 0}$, $\forall k \in \mathbb{Z}$ 且 $0 \leq k \leq n$, 有:

$$\binom{n}{k} = \frac{n \times (n-1) \times \cdots \times (n-k+1)}{k \times (k-1) \times \cdots \times 1} \quad \text{且} \quad \binom{n}{k} = \frac{n!}{k! (n-k)!}$$

Proof: 第一个位置有 n 种选法. 第~~二~~²个位置有 $(n-1)$ 种选法.

$\dots$ 第 k 个位置有 $n-(k-1) = n-k+1$ 种选法.

$k!$ 个排列是相同的子集.

$$\therefore \binom{n}{k} = \frac{n \times (n-1) \times \cdots \times (n-k+1)}{k \times (k-1) \times \cdots \times 1} = \frac{n!}{k! (n-k)!} \quad \square$$

定理 (Kummer's theorem) p 是素数, $n \in \mathbb{Z}_{\geq 0}$, $m \in \mathbb{Z}$ 且 $0 \leq m \leq n$.

对 $\forall x \in \mathbb{Z}_{\geq 0}$, 已经证明了存在唯一的 ~~展开~~ p 进制展开:

$$x = a_0 + a_1 p + a_2 p^2 + \dots + a_r p^r \quad (\text{其中 } r \in \mathbb{Z}_{\geq 0}, a_i \in \{0, 1, \dots, p-1\})$$

记 $S_p(x) = \sum_{i=0}^r a_i$. 则有:

$$v_p \left(\binom{n}{m} \right) = \frac{S_p(m) + S_p(n-m) - S_p(n)}{p-1}$$

= 在 p 进制下将 m 与 $n-m$ 相加产生的进位总次数.

Proof: $\because n \in \mathbb{Z}_{\geq 0}$ 且 $m \in \mathbb{Z}$ 且 $0 \leq m \leq n$

$$\therefore \binom{n}{m} = \frac{n!}{m!(n-m)!} \quad \because \binom{n}{m} \in \mathbb{Z}_{\geq 1} \quad \therefore m!(n-m)! \mid n!$$

$$\therefore v_p \left(\binom{n}{m} \right) = v_p \left(\frac{n!}{m!(n-m)!} \right) = v_p(n!) - v_p(m!(n-m)!)$$

$$= v_p(n!) - v_p(m!) - v_p((n-m)!)$$

$$= \frac{n - S_p(n)}{p-1} - \frac{m - S_p(m)}{p-1} - \frac{(n-m) - S_p(n-m)}{p-1}$$

$$= \frac{n - S_p(n) - m + S_p(m) - n + m + S_p(n-m)}{p-1} = \frac{S_p(m) + S_p(n-m) - S_p(n)}{p-1}$$

显然 $m + (n - m) = n$.

$$\text{设 } m = x_0 + x_1 p + x_2 p^2 + \dots + x_N p^N \quad (x_i \in \{0, 1, \dots, p-1\})$$

$$n - m = y_0 + y_1 p + y_2 p^2 + \dots + y_N p^N \quad (y_i \in \{0, 1, \dots, p-1\})$$

$$n = z_0 + z_1 p + z_2 p^2 + \dots + z_N p^N \quad (z_i \in \{0, 1, \dots, p-1\})$$

$$\text{则 } m = x_N p^N + \dots + x_2 p^2 + x_1 p + x_0$$

$$n - m = y_N p^N + \dots + y_2 p^2 + y_1 p + y_0$$

$$n = z_N p^N + \dots + z_2 p^2 + z_1 p + z_0 \quad (\text{可以取充分大的 } N, \text{ 顶部补 } 0)$$

记 C_i 为第 i 位向第 $i+1$ 位进位的值 (i 从 0 开始) (记 $C_{-1} = 0$)

$$\therefore C_i \in \{0, 1\}$$

$\therefore$ 逐位列出竖式加法:

$$\text{第 } 0 \text{ 位: } x_0 + y_0 + C_{-1} = C_0 \cdot p + z_0$$

$$\text{第 } 1 \text{ 位: } x_1 + y_1 + C_0 = C_1 \cdot p + z_1$$

$$\text{第 } 2 \text{ 位: } x_2 + y_2 + C_1 = C_2 \cdot p + z_2$$

...

$$\text{第 } (N-1) \text{ 位: } x_{N-1} + y_{N-1} + C_{N-2} = C_{N-1} \cdot p + z_{N-1}$$

$$\text{第 } N \text{ 位: } x_N + y_N + C_{N-1} = C_N \cdot p + z_N$$

$$\therefore \left(\sum_{i=0}^N x_i \right) + \left(\sum_{i=0}^N y_i \right) + \left(\sum_{i=0}^{N-1} C_i \right) = \left(\sum_{i=0}^N C_i \right) p + \left(\sum_{i=0}^N z_i \right)$$

分两种情况讨论：

(i) 如果 $C_N = 0$ ，则有：

$$S_p(m) + S_p(n-m) + \left(\sum_{i=0}^{N-1} C_i \right) = \left(\sum_{i=0}^{N-1} C_i \right) p + S_p(n)$$

$$\therefore S_p(m) + S_p(n-m) - S_p(n) = \left(\sum_{i=0}^{N-1} C_i \right) (p-1)$$

$$\therefore \frac{S_p(m) + S_p(n-m) - S_p(n)}{p-1} = \sum_{i=0}^{N-1} C_i = \text{在 } p \text{ 进制下将 } m \text{ 与 } n-m \text{ 相加产生的进位总次数.}$$

(ii) 如果 $C_N = 1$ ，则有：

$$\left(\sum_{i=0}^N x_i \right) + \left(\sum_{i=0}^N y_i \right) + \left(\sum_{i=0}^N C_i \right) = \left(\sum_{i=0}^N C_i \right) p + \left(\sum_{i=0}^N z_i + C_N \right)$$

$$S_p(m) + S_p(n-m) + \left(\sum_{i=0}^N C_i \right) = \left(\sum_{i=0}^N C_i \right) p + S_p(n)$$

$$\therefore S_p(m) + S_p(n-m) - S_p(n) = \left(\sum_{i=0}^N C_i \right) (p-1)$$

$$\therefore \frac{S_p(m) + S_p(n-m) - S_p(n)}{p-1} = \sum_{i=0}^N C_i = \text{在 } p \text{ 进制下将 } m \text{ 与 } n-m \text{ 相加产生的进位总次数.}$$

